Detection of Fake vs Genuine Skills in Resumes using NLP, ATS and Contextual Analysis

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Abstract
Due to the highly competitive nature of jobs today, many recruitment professionals face difficulties when analysing thousands of resumes for a given vacancy due to inefficiency, discrimination, and loss of potential talent. Existing approaches for screening applicants, namely manual resume analysis and ATS algorithms, are inefficient, non-scalable and subject to human bias and mistakes. Therefore, this paper aims at developing a new intelligent, automated system for resume analysis that employs technologies such as Python programming language, NLP, ML, and ATS algorithms. The developed tool will analyse unstructured resumes and extract essential information about each candidate such as personal information, education, work experience, acquired skills and certifications. One of the inventions incorporated in this system is project-experience based skills matching. Through applying advanced NLP techniques, textual information from resumes will be processed into meaningful structures that can then be compared with given job requirements. Experiments conducted on real datasets have shown very high rates of efficiency in terms of resume parsing, candidate scoring, and job requirement fit analysis.

Keywords— Resume Screening, Natural Language Processing (NLP), Machine Learning, Applicant Tracking System (ATS), Resume Parsing, Candidate Ranking, Job Description Matching, Skills Extraction, Keyword Matching, Automated Resume Evaluation, Deep Learning, Text Mining, Candidate Profile Analysis, Recruitment Automation, Job Fit Analysis, Intelligent Filtering

I. INTRODUCTION

The recruitment ecosystem has undergone significant transformation with the integration of Artificial Intelligence (AI). Organizations increasingly rely on online recruitment platforms such as LinkedIn, Naukri, and Indeed to source candidates. With the rapid digitization of hiring processes, the volume of incoming resumes has grown exponentially, making manual screening inefficient and time-consuming. Despite its advantages, the adoption of automated systems has introduced new challenges. One major issue is the manipulation of resumes, where candidates exaggerate or falsify their skills to pass Applicant Tracking System (ATS) screening processes. Such practices can lead to the selection of unqualified candidates, resulting in wasted time, resources, and reduced organizational productivity. Therefore, there is a need for a system that not only extracts information from resumes but also validates the authenticity of the claimed skills. Current ATS-based systems primarily rely on keyword-matching techniques to filter applications, but they fail to verify whether candidates genuinely possess the listed skills. Studies such as those by [1]Yadav et al. demonstrate the effectiveness of Natural Language Processing (NLP) in extracting structured information from resumes. However, these approaches still depend heavily on keyword matching without incorporating validation mechanisms.

This limitation motivates the development of a system that detects fake or exaggerated skills using contextual understanding and project-based validation techniques.

II. PROBLEM STATEMENT

Current resume screening systems suffer from several limitations that affect the quality of candidate selection.

- Excessive reliance on keyword matching.
- Lack of contextual understanding of skills.
- Absence of validation between claimed skills and practical experience.
- High rate of false positives in candidate selection.

Existing ATS-based systems evaluate resumes based on parameters such as keyword frequency, number of skills, and years of experience. However, they do not verify the authenticity or relevance of these parameters.

As a result, candidates without the required expertise may be selected, leading to inefficiencies in the recruitment process. Additionally, these systems lack transparency, as they do not provide clear reasoning behind candidate selection or rejection.

Moreover, candidates with strong practical knowledge but fewer keywords may be overlooked, while those with keyword-rich resumes but limited experience may be prioritized.

III. RELATED WORK

Recent advancements in Natural Language Processing (NLP) and machine learning have significantly improved resume processing and candidate-job matching. Techniques such as BERT and Word2Vec embeddings have enabled a transition from traditional keyword-based approaches to semantic understanding of resumes and job descriptions.

Several researchers have proposed effective methodologies in this domain. Yadav et al. introduced an NLP-based ATS system for extracting and ranking resume data. Liang et al. utilized deep learning techniques to enhance entity recognition and segmentation in resumes. Bharadwaj et al. developed a resume classification and job-matching system using LSTM-based neural networks.

In parallel, fraud detection techniques in domains such as finance, e-commerce, and social media have leveraged anomaly detection, behavioral analysis, and inconsistency identification. Knowledge graph-based approaches have also been used to model relationships between skills, technologies, and projects, enabling logical validation of candidate claims.

Despite these advancements, most existing systems focus on resume parsing, information extraction, and ranking. There is limited research on detecting falsified or exaggerated skills, highlighting a critical research gap.

IV. OBJECTIVE

The primary objective of this study is to develop an intelligent system capable of verifying the authenticity of skills claimed by job applicants.

The key objectives are as follows:

- To design and implement an NLP and machine learning-based approach for identifying skills and experiences in resumes.
- To develop a semantic verification module that analyzes the logical consistency between claimed skills and supporting evidence.
- To build a fraud detection mechanism to identify anomalies and inconsistencies in resumes.
- To utilize knowledge graphs for establishing relationships between skills, technologies, and projects.
- To evaluate system performance using metrics such as accuracy, precision, recall, and F1-score.

V. PROPOSED METHODOLOGY

The proposed system detects fake or exaggerated skills resumes through a structured multi-stage approach.

- Data Collection: Resumes are collected from Kaggle datasets and online job portals across domains such as IT, engineering, and management. Both genuine and modified resumes with injected fake skills are included.
- Data Preprocessing: The collected text is cleaned by converting it to lowercase, removing stop words, eliminating special characters, and normalizing spacing.
- Information Extraction: Important information such as skills, education, work experience, projects, and certifications is extracted using NLP techniques.
- Keyword and Skill Matching: Extracted skills are compared with job descriptions to evaluate initial relevance.
- Contextual Analysis: The system evaluates how skills are described. Detailed descriptions indicate authenticity, whereas vague mentions are flagged as suspicious.
- Project-Skill Validation: This core component verifies whether claimed skills are supported by project experience. Skills without supporting evidence are flagged.
- Consistency Analysis: The system checks alignment between skills and other resume sections, including work experience, education, and certifications.
- Scoring Mechanism: A weighted scoring model is used as follows:

  Final Score = (Contextual Analysis × 0.4) + (Project-Skill Validation × 0.3) + (Consistency Analysis × 0.2) + (Keyword Matching × 0.1)

- Classification: Based on the final score, resumes are classified as Genuine (Score ≥ 0.75), Suspicious ($0.45 \leq$ Score < 0.75), or Fake/Exaggerated (Score < 0.45).

VI. SYSTEM ARCHITECTURE

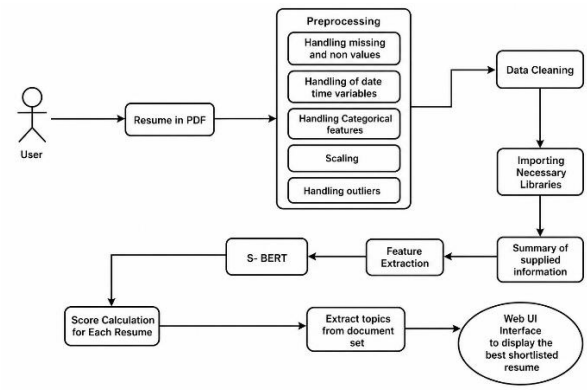

Figure 1Architecture of automatic review of resumes

Fig. 1 illustrates the architecture of the Automatic Resume Review system.

As shown in Fig. 1, the process consists of five major stages that together form the complete automated resume review pipeline. Each stage plays a crucial role in ensuring accurate and efficient candidate evaluation.

1. Data Collection: Resumes are collected from various sources such as job portals, career websites, and corporate platforms. Along with resumes, job descriptions and role requirements are also gathered for comparison.
2. Preprocessing: In this stage, irrelevant data such as stop words, punctuation marks, and noise are removed from both resumes and job descriptions. Techniques such as tokenization, stemming, and lemmatization are applied to standardize the text.
3. Feature Extraction: Important features are extracted from the preprocessed data using Natural Language Processing (NLP) techniques such as Sentence-BERT (S-BERT). These features are converted into embeddings that capture semantic similarity and contextual meaning.
4. Scoring: A ranking score is calculated for each candidate based on cosine similarity between the resume and the job description. Candidates with higher similarity scores are ranked higher.
5. Candidate Rejection: Candidates who do not meet the minimum threshold of cosine similarity are rejected or placed in a low-priority list for manual review.

VII. RESULTS AND ANALYSIS

The project is currently at the concept and prototype stage. However, preliminary analysis indicates several key observations. Existing Applicant Tracking Systems (ATS) primarily rely on keyword-based search techniques, which often overlook context and semantic meaning.

In contrast, the proposed system integrates multiple validation stages to improve the accuracy and reliability of candidate screening.

The projected results include lower false positives, enhanced screening of candidates, and increased efficiency in the recruitment process.

Unlike traditional ATS scoring methods based on word count, skills, and experience, this system ensures that extracted skills are verified for authenticity and contextual relevance.

VIII. DISCUSSION

The combination of Natural Language Processing (NLP) and context validation significantly enhances the quality of resume screening. This study demonstrates that keywords alone are not sufficient for accurate evaluation.

Context validation plays a crucial role in verifying candidate information and ensuring meaningful matching. Furthermore, even simple NLP techniques can outperform traditional ATS methods by improving accuracy and reducing irrelevant matches.

IX. APPLICATIONS

The proposed system can be effectively applied in various domains, including corporate hiring systems, campus placements, freelancing platforms, and recruitment agencies. These applications highlight its potential to improve efficiency and reliability in candidate selection processes.

X. FUTURE WORK

Future enhancements of the system may include integration with GitHub for code validation, LinkedIn profile analysis for professional verification, and the use of advanced deep learning models such as BERT and Large Language Models (LLMs). Additionally, implementing real-time resume scoring systems can further improve the responsiveness and scalability of the solution.

XI. CONCLUSION

This research presents an improved framework for detecting fake versus genuine skills in resumes. By combining ATS algorithms with NLP-based contextual and project validation, the proposed system effectively addresses key limitations in existing recruitment processes. It provides a more reliable and efficient approach to candidate evaluation and demonstrates strong potential for real-world implementation.

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